RMO– 2015(2)

Max Mark – 102 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions. All Questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. Let ABC be a triangle. Let B' and C' denote respectively the reflection of B and C in the internal angle bisector of $\angle A$. Show that the triangle ABC and $AB'C'$ have the same incentre.
2. Let $P(x)=x^2+ax+b$ be quadratic polynomial with real coefficients. Suppose there are real numbers $s \neq t$ such that $P(s)=t$ and $P(t)=s$. Prove that $b-st$ is root of the equation $x^2+ax+b-st=0$.
3. Find all integers a, b, c such that $a^2=bc+1, \quad b^2=ca+1$.
4. Suppose 32 objects are placed along a circle at equal distances. In how many ways can 3 objects be chosen from among them so that no two of the three chosen objects are adjacent nor diametrically opposite?
5. Two circles Γ and Σ in the plane intersect at two distinct points A and B, and the centre of Σ lies on Γ. Let points C and D be on Γ and Σ, respectively, such that C, B and D are collinear. Let point E on Σ be such that DE is parallel to AC. Show that AE = AB.
6. Find all real number a such that $4<a<5$ and $a(a-3\{a\})$ is an integer. (here {a} denotes the fractional part of a. For example {1.5}=0.5, {-3.4}=0.6.)

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